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ChE 210 Exam #1 19 February 2026

To receive full credit, you must show all your work.

(1) (55 pts.) Hydrogen chloride (HCl) and oxygen (O_2) are reacted in the gas phase to form chlorine (Cl_2) and water (H_2O). A gas feed stream entering the reactor contains 100 mol/min of HCl and air (79 mol% N_2, 21 mol% O_2). The air in the stream contains 30% more O_2 than stoichiometrically required to oxidize all the HCl. In the reactor, 60% of the HCl is oxidized. The product stream contains all the Cl_2 and H_2O produced, the inert N_2, and the unreacted portion of the HCl and O_2.

(a) Write a balanced chemical reaction equation for this reaction. Use any method you like to balance the equation. (5 pts.)

$$4\,HCl + 1\,O_2 + 1\,N_2 \longrightarrow 2\,Cl_2 + 2\,H_2O + 1\,N_2$$

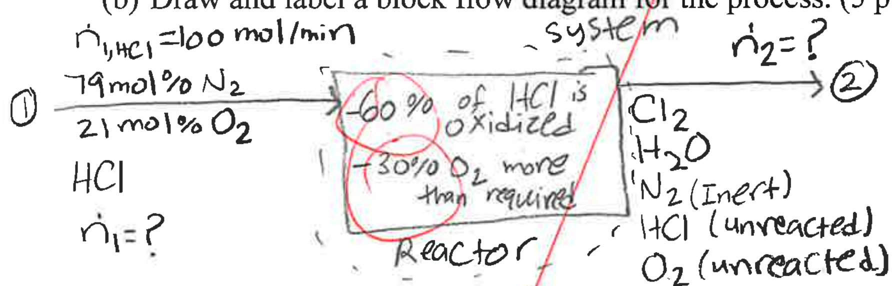
The HCl and O_2 that didn't react are not part of the equation I would assume

N_2 is inert and should not be in Rx eqn.

True.

4

(b) Draw and label a block flow diagram for the process. (5 pts.)



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(c) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. Give values for 4a, 4b, 5a, 5b, 5c, and 5d. Compute the DOF. (5 pts.)

4a) 8
4b) 1
9

5a) 1
5b) 1
5c) 2
5d) 5
9

$\text{DoF} = 9 - 9 = \boxed{0 \text{ well specified}}$

5

(d) Calculate the molar flow rates of the O_2 and N_2 in the feed stream to the reactor. (10 pts.)

N_2:

$\cancel{100\,\frac{\text{mol HCl}}{\text{min}} \times \frac{1 \text{ mol } N_2}{4 \text{ mol HCl}} = 25 \text{ mol } N_2/\text{min}}$

No! N_2 doesn't react.

O_2:

$100\,\frac{\text{mol HCl}}{\text{min}} \times \frac{1 \text{ mol } O_2}{4 \text{ mol HCl}} = 25 \text{ mol } O_2/\text{min} \rightarrow \text{total}$

3

$(25)(0.30) = 7.5$

$25 - 7.5 = 17.5\ \frac{\text{mol } O_2}{\text{min}}$

plus

reacts

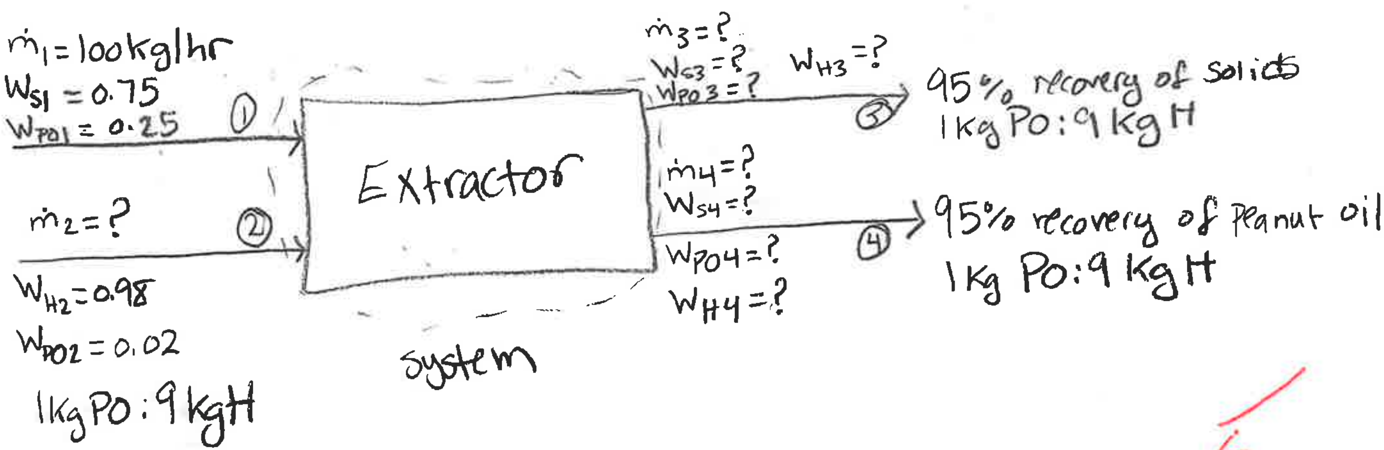
Feed Stream molar flow rate:

Acid (HCl) (mol/min)	Oxygen (O_2) (mol/min)	Nitrogen (N_2) (mol/min)
100 given	~~25~~	~~25~~

(2) (45 pts.) Hexane can be used to extract peanut oil from ground peanuts. There are two streams entering the extractor unit: (1) the feed stream $\dot{m}_1 = 100$ kg/hr containing ground peanuts which has a composition of 75 wt% solids (S) and 25 wt% peanut oil (PO), and (2) a stream $\dot{m}_2$ of 98 wt% hexane (H) and 2 wt% peanut oil. There are two streams exiting the extractor unit: (3) a waste stream $\dot{m}_3$ containing solids, peanut oil and hexane, and (4) the product stream $\dot{m}_4$ which also contains solids, peanut oil and hexane. The waste stream recovers 95% of the solids entering the extractor while the product stream recovers 95% of the peanut oil entering the extractor. The equilibrium solubility of peanut oil in hexane is 1 kg of peanut oil to 9 kg of hexane. Assume that this equilibrium has been achieved in both exit streams.

(a) Draw a block flow diagram of the process and label all the streams. (5 pts.)

$* \; W_{i\#} =$ weight %



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(b) Solve the appropriate mass balances along with the recovery equation above to calculate all the species and total mass flow rates in kg/hr. Fill in the table below with your answers. (40 pts.)

$\cancel{Acc.} = In - Out + \cancel{gen} - \cancel{Cons.} \rightarrow In = out$

S.S.   no rxn

9 unknowns
need 9 equations

① $\boxed{\dot{m}_1 + \dot{m}_2 = \dot{m}_3 + \dot{m}_4}$

② $\boxed{\dot{m}_1 W_{S1} = \dot{m}_3 W_{S3} + \dot{m}_4 W_{S4}} \Rightarrow \dot{m}_{1S} = \dot{m}_{3S} + \dot{m}_{4S}$

10

③ $\boxed{\dot{m}_1 W_{PO1} + \dot{m}_2 W_{PO2} = \dot{m}_3 W_{PO3} + \dot{m}_4 W_{PO4}} \Rightarrow \dot{m}_{1PO} + \dot{m}_{2PO} = \dot{m}_{3PO} + \dot{m}_{4PO}$

④ $\boxed{\dot{m}_2 W_{H2} = \dot{m}_3 W_{H3} + \dot{m}_4 W_{H4}} \Rightarrow \dot{m}_{2H} = \dot{m}_{3H} + \dot{m}_{4H}$

⑤ $\boxed{0.95 W_{S1} = W_{S3}} \Rightarrow$ 95% of what goes in will go out

⑥ $\boxed{0.95(W_{PO1} + W_{PO2}) = W_{PO4}} \Rightarrow$ 95% of what goes in will go out

10

⑦ $\boxed{0.05 W_{S1} = W_{S4}} \Rightarrow$ 5% of what goes in will go out

⑧ $\boxed{0.05(W_{PO1} + W_{PO2}) = W_{PO3}} \Rightarrow$ 5% of what goes in will go out

Problem # 2 Cont. NAME: Lama Alsurakhi

$$\frac{(100)(0.75)}{0.7125} - \frac{\dot{m}_4(0.0375)}{0.7125} + \dot{m}_4 - 100 = \frac{(100)(0.75)}{0.7125} - \frac{\dot{m}_4(0.0375)}{0.7125} + \dot{m}_4(0.706)$$

you are confusing w and m throughout.

$$\dot{m}_4 - 100 = \dot{m}_4(0.706)$$

$$0.294\dot{m}_4 = 100 \quad , \quad \dot{m}_4 = 340.136 \text{ kg/hr}$$

Species n in stream 4:

S:

$$340.136 \frac{\text{kg tot}}{\text{hr}} \times \frac{0.0375 \text{kg S}}{1 \text{kg tot}} = 12.7551 \text{ kg S/hr}$$

Po:

$$340.136 \frac{\text{kg tot}}{\text{hr}} \times \frac{0.2565 \text{kg Po}}{1 \text{kg tot}} = 87.2449 \text{ kg Po/hr}$$

H:

$$340.136 \frac{\text{kg tot}}{\text{hr}} \times \frac{0.706 \text{ kg H}}{1 \text{kg tot}} = 240.136 \text{ kg H/hr}$$

Finding $\dot{m}_3$: eqn. 2

$$\dot{m}_3 = \frac{(100)(0.75) - (340.136)(0.0375)}{0.7125} = 87.3613 \text{ kg/hr}$$

Species n in Stream 3:

S:

$$87.3613 \frac{\text{kg tot}}{\text{hr}} \times \frac{0.7125 \text{kg S}}{1 \text{kg tot}} = 62.2449 \text{ kg S/hr}$$

Po:

$$87.3613 \frac{\text{kg tot}}{\text{hr}} \times \frac{0.0135 \text{kg Po}}{1 \text{kg tot}} = 1.17938 \text{ kg Po/hr}$$

H:

$$87.3613 \frac{\text{kg tot}}{\text{hr}} \times \frac{0.274 \text{kg H}}{1 \text{kg tot}} = 23.937 \text{ kg H/hr}$$

2 Cont. →

Finding $\dot{m}_2$: eqn. 1

$$\dot{m}_2 = 87.3613 + 340.136 - 100 = 327.497 \text{ kg/hr}$$

Finding species $\dot{n}$ in Stream 2

Po:

$$327.497 \frac{\text{kg} \cancel{\text{tot}}}{\text{hr}} \times \frac{0.02 \text{ kg Po}}{1 \text{ kg} \cancel{\text{tot}}} = 6.54995 \text{ kg Po/hr}$$

H:

$$327.497 \frac{\cancel{\text{kg tot}}}{\text{hr}} \times \frac{0.98 \text{ kg H}}{\cancel{1 \text{ kg tot}}} = 320.947 \text{ kg H/hr}$$

(Problem 1 Continued) NAME Lama Alsuvakhi

(e) Calculate the species molar flow rates for all five components in the product stream in mol/min. Show all your calculations and write your final answers in the space provided below. (30 pts.)

ss.s.

$\cancel{Acc.} = In - Out + gen - Cons$

$0 = In - out + gen - Cons$

1	39
2	28
T	67

HCl:

$0 = \dot{n}_{HCl,1} - \dot{n}_{HCl,2} + 0 - \dot{n}_{HCl,cons}$

$\dot{n}_{HCl_{cons}} = (0.6)\dot{n}_{HCl,1} = (0.6)(100 \frac{mol}{min}) = 60 \frac{mol}{min}$ HCl, cons

$\dot{n}_{HCl_2} = \dot{n}_{HCl,1} - \dot{n}_{HCl,cons} \Rightarrow 100 - 60 = 40 \frac{mol}{min}$ HCl, 2

N_2:

$\dot{n}_{N_2,1} = \dot{n}_{N_2,2} \Rightarrow 25 \frac{mol}{min}$ $N_{2,2}$

O_2: 30% more than needed

How much extra O_2 : $(25 \frac{mol}{min})(0.30) = 7.5$ ← unreacted O_2 ok

$\dot{n}_{O_2,cons} = 25 - 7.5 = 17.5 \frac{mol}{min}$ O_2, cons. → reacted

$\dot{n}_{O_2,2} = 25 - 17.5 = 7.2 \frac{mol}{min}$ O_2, 2 math?

Cl_2: gen: $2 \times \dot{n}_{HCl,cons} = \cancel{2}(60) = 1\cancel{20} \frac{mol}{min}$ Cl_2, gen

$\dot{n}_{Cl_2,gen} = \dot{n}_{Cl_2,2} \Rightarrow \cancel{120 \frac{mol}{min}}$ Cl_2, 2

H_2O: gen: $2 \times \dot{n}_{HCl,Cons} = \cancel{2}(60) = 12\cancel{0} \frac{mol}{min}$ H_2O, gen

$\dot{n}_{H_2O,gen} = \dot{n}_{H_2O,2} \Rightarrow 120 \frac{mol}{min}$ $\cancel{H_2O}$, 2 ZZ

Product Stream molar flow rate:

Acid (HCl) (mol/min)	Oxygen (O_2) (mol/min)	Nitrogen (N_2) (mol/min)	Chlorine (Cl_2) (mol/min)	Water (H_2O) (mol/min)
40	7.5	25	~~120~~	~~120~~

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(Problem 2 Continued)

Stream ①

PO:

$\frac{100\ \cancel{kg\ tot}}{hr} \times \frac{0.25\ kg\ PO}{1\ \cancel{kg\ tot}} = 25\ kg\ PO/hr$

S:

$\frac{100\ \cancel{kg\ tot}}{hr} \times \frac{0.75\ kg\ S}{1\ \cancel{kg\ tot}} = 75\ kg\ S/hr$

Using eq. 5–8 to find wt% of S and PO in streams 3 and 4

5) $0.95(0.75) = 0.7125 \Rightarrow W_{S3}$

7) $0.05(0.75) = 0.0375 \Rightarrow W_{S4}$

× 100 kg/hr

not mass fractions

6) $0.95(0.25 + 0.02) = 0.2565 \Rightarrow W_{PO4}$ 100

8) $0.05(0.25 + 0.02) = 0.0135 \Rightarrow W_{PO3}$ 100

Find wt% of hexane in Streams 3 and 4

Stream 3: $W_{H3} = 1 - 0.7125 - 0.0135 = 0.274 \Rightarrow W_{H3}$?

3

Stream 4: $W_{H4} = 1 - 0.0375 - 0.2565 = 0.706 \Rightarrow W_{H4}$

Solving for $\dot{m}_4$: Using eq. 1, 2, and 4

① $\dot{m}_2 = \dot{m}_3 + \dot{m}_4 - \dot{m}_1$ ② $\dot{m}_3 = \frac{\dot{m}_1 W_{S1} - \dot{m}_4 W_{S4}}{W_{S3}}$ Plug into ④

$(\dot{m}_3 + \dot{m}_4 - \dot{m}_1) = \frac{\dot{m}_1 W_{S1} - \dot{m}_4 W_{S4}}{W_{S3}} + \dot{m}_4 W_{H4}$

$\frac{\dot{m}_1 W_{S1} - \dot{m}_4 W_{S4}}{W_{S3}} + \dot{m}_4 - \dot{m}_1 = \frac{\dot{m}_1 W_{S1} - \dot{m}_4 W_{S4}}{W_{S3}} + \dot{m}_4 W_{H4}$

Next sheet

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Species∵Stream#	1	2	3	4
PO (kg PO/hr)	25	6.550	62.2449	87.2449
S (kg S/hr)	75	0	1.1794	12.7551
H (kg H/hr)	0	320.947	23.937	240.136
Total (kg/hr)	100	327.497	87.3613	340.136

* Numbers are rounded